Quantum Topology-GRPO (Revised Version) A Variational Quantum Hierarchical Reinforcement Learning Algorithm with Global Topological Calibration for Intra-Graph Inter-Group Scenarios

Lanhaijian
Hongqiao University

Abstract

This paper proposes Quantum Topology-GRPO, a fully quantum-enhanced extension of the revised Topology-GRPO framework for graph-structured decision systems. The design strictly adheres to the core GRPO principle: no intra-group critics, advantage computed purely from normalized rewards, with a global quantum value function reserved exclusively for topological calibration. All classical modules are replaced with quantum-native implementations: inter-group communication via Bell-pair entanglement and quantum teleportation, topological advantage accelerated by quantum amplitude estimation, inter-group consistency enforced via SWAP-test state fidelity, and policy stability guaranteed by von Neumann entropy regularization.

The macro effective duality coupling $\mathcal{C}^q_{\text{eff}}$ and q-deformed Q-Fuse are philosophically inspired by Self-Entangled Duality Dynamics (SED), but no rigorous mathematical derivation connects SED's quantum spacetime algebra to the present macroscopic engineering framework. All technical components are independent assumptions validated by task performance.

1 Introduction

The revised Topology-GRPO framework eliminates intra-group critics and relies on reward normalization for advantage estimation, with a global value function retained only for cross-group topological validation. While this design achieves cleaner objectives and more stable training, classical implementations still face fundamental limitations in high-dimensional graph scenarios: limited expressivity for correlated group behaviors, linear scaling with neighborhood size, and inefficient global state representation.

Quantum computing provides a natural solution to these limitations:

- Quantum superposition enables exponential state representation and parallel evaluation.
- Quantum entanglement natively models inter-group correlations without explicit message overhead.
- Quantum amplitude estimation provides quadratic speedup for neighborhood reward statistics.
- Quantum state fidelity directly measures inter-group policy alignment without classical distance computation.

This paper upgrades the framework to Quantum Topology-GRPO with:

1. Fully quantum policy networks and message passing, with no intra-group critics.
2. Bell-pair entanglement and quantum teleportation for inter-group communication.
3. Quantum amplitude estimation for accelerated three-component topological advantage.
4. Global quantum value function V^q_{global} for topological calibration only.
5. SWAP-test fidelity constraints and von Neumann entropy regularization for policy stability.

2 Preliminaries and Quantum Formulation

We model the system as a weighted graph:

$$G = (V, E, W), \quad V = \bigcup_{g=1}^{k} V_g, \quad E = E_{\text{intra}} \cup E_{\text{inter}}. \tag{1}$$

In the quantum formulation:

- Nodes and groups are encoded into separable quantum registers $|\psi_g\rangle$.
- Inter-group edges are represented by pre-distributed Bell pairs:

$$|\Phi^+\rangle_{gh} = \frac{1}{\sqrt{2}} \left(|00\rangle_{gh} + |11\rangle_{gh}\right). \tag{2}$$

- The optimization objective is to maximize the expected discounted return:

$$\max_{\pi_h^q, \{\pi_l^{g,q}\}} \mathbb{E}\left[\sum_{g,t} \gamma^t R^{g,q}\right] \quad \text{s.t. inter-group quantum policy consistency.} \tag{3}$$

3 Quantum Topology-GRPO Framework

3.1 Core Design Principles

1. No intra-group quantum critics: no local value functions within any group.
2. Advantage from quantum reward statistics only: no TD learning, no local value fitting.
3. Global quantum value for topological calibration only: V_{global}^q exclusively for cross-group alignment.
4. Quantum-native modules: all components implemented with parameterized quantum circuits (PQCs).

3.2 Quantum State Encoding

Classical group states are mapped to quantum states via amplitude encoding:

$$|\tilde{\psi}_g\rangle = U_{\text{enc}}(s_g) \otimes U_{\text{msg}}(m_g)|0\rangle^{\otimes n}, \tag{4}$$

where U_{enc} performs amplitude embedding for state information, U_{msg} performs phase encoding for inter-group messages, and n is the total number of qubits per group register.

3.3 Bell-Pair Quantum Inter-Group Message Passing

Inter-group communication is implemented via quantum teleportation over pre-distributed Bell pairs:

$$|\psi_{\text{msg}}\rangle_g = \text{Tr}_h \left[U_{gh}(\theta_{gh})|\tilde{\psi}_g\rangle \otimes |\tilde{\psi}_h\rangle \otimes |\Phi^+\rangle_{gh}\right], \tag{5}$$

where $U_{gh}(\theta_{gh})$ is a parameterized two-qubit gate controlled by the topological coupling weight W_{gh}.

The received message is fused with the local state via quantum gating:

$$|\tilde{\psi}_g^{\text{enhanced}}\rangle = \text{QGateFuse}\left(|\tilde{\psi}_g\rangle, |\psi_{\text{msg}}\rangle_g\right), \tag{6}$$

implemented with RY rotation gates and quantum attention weighting.

3.4 Global Quantum Value for Topological Calibration

A global quantum value function is defined exclusively for cross-group credit alignment and topological validation:

$$V_{\text{global}}^q = \langle\psi_{\text{global}}|U_q^\dagger Z^{\otimes k} U_q|\psi_{\text{global}}\rangle, \tag{7}$$

where $|\psi_{\text{global}}\rangle$ is the joint global quantum state, and $Z^{\otimes k}$ is the multi-qubit Pauli-Z measurement operator. This value is not a learned critic and is only used to stabilize global advantage normalization.

3.5 Quantum Topological Advantage with Amplitude Estimation

The three-component advantage is computed purely from quantum reward statistics, accelerated by quantum amplitude estimation (Grover-type quadratic speedup from $O(|\mathcal{N}(g)|)$ to $O(\sqrt{|\mathcal{N}(g)|})$):

$$\begin{aligned}\hat{A}^{g}_{i,\text{quantum}} =& \alpha \cdot \frac{R_i^{\text{global}} - V^q_{\text{global}}}{\sigma^{\text{global}}} \\ &+ (1-\alpha) \cdot \frac{r_i^g - \bar{r}^g}{\sigma^g} \\ &+ \beta \cdot \frac{r_i^g - \bar{r}^{\mathcal{N}(g)}}{\sigma^{\mathcal{N}(g)}}.\end{aligned} \tag{8}$$

All terms are derived from normalized rewards, with no intra-group value functions involved.

3.6 Macro Effective Quantum Duality Coupling $\mathcal{C}^q_{\text{eff}}$

Motivation In quantum-enhanced graph systems, inter-group entanglement and decoherence dynamics create two distinct regimes: weak quantum coupling where classical-quantum hybrid fusion suffices, and strong coherent coupling where full q-deformed non-commutative fusion is required.

Remark The macro effective duality coupling $\mathcal{C}^q_{\text{eff}}$ shares conceptual inspiration with the duality coupling $\mathcal{C}$ in Self-Entangled Duality Dynamics (SED). However, $\mathcal{C}^q_{\text{eff}}$ is an independent engineering parameter for quantum graph dynamics. There is no strict mathematical derivation between SED's quantum spacetime algebra and the present framework. Their mathematical domains, physical dimensions, and research objects are fundamentally different.

Definition Let F_{gh} be the Bell-pair fidelity between groups g and h, $T_1^{(g)}, T_2^{(g)}$ denote the coherence and decoherence time of group g's quantum register, τ_{gate} the average two-qubit gate execution time, and $\text{Var}_q(r_h)$ the quantum-enhanced reward variance estimated via amplitude estimation. $\sigma^{\mathcal{N}(g)}$ is the neighbor reward standard deviation from Eq. (8), and $\epsilon = 10^{-8}$ a numerical stabilizer. The normalized macro effective quantum duality coupling is formulated as:

$$\mathcal{C}^q_{\text{eff}}{}^{(g)} = \frac{\min(T_1^{(g)}, T_2^{(g)})}{\tau_{\text{gate}} \cdot |\mathcal{N}(g)|} \sum_{h \in \mathcal{N}(g)} F_{gh} \cdot \frac{\text{Var}_q(r_h)}{(\sigma^{\mathcal{N}(g)})^2 + \epsilon}. \tag{9}$$

Critical Threshold and Operating Regimes The critical threshold $\mathcal{C}^q_c$ can be set as a fixed hyperparameter or optimized as a learnable parameter during training. Two operating regimes are divided by $\mathcal{C}^q_c$:

- $\mathcal{C}^q_{\text{eff}}{}^{(g)} < \mathcal{C}^q_c$: **Weak quantum coupling regime**. Adopt classical-quantum hybrid CONCAT+RY fusion.
- $\mathcal{C}^q_{\text{eff}}{}^{(g)} \geq \mathcal{C}^q_c$: **Strong quantum coherent regime**. Activate the q-deformed Q-Fuse$_q$ fusion module.

3.7 Adaptive Quantum Fusion Operator

The adaptive fusion rule is given by:

$$|\tilde{\psi}_g^{\text{enhanced}}\rangle = \begin{cases} \text{QGateFuse}_{\text{hybrid}}\left(|\tilde{\psi}_g\rangle, |\psi_{\text{msg}}\rangle_g\right) & \mathcal{C}^q_{\text{eff}}{}^{(g)} < \mathcal{C}^q_c \\ \text{Q-Fuse}_q\left(|\tilde{\psi}_g\rangle, |\psi_{\text{msg}}\rangle_g; \tilde{\theta}\right) & \mathcal{C}^q_{\text{eff}}{}^{(g)} \geq \mathcal{C}^q_c \end{cases} \tag{10}$$

Q-Fuse$_q$: Quantum q-Deformed Non-Commutative Fusion Implemented as a parameterized quantum circuit with ZZ-type Ising coupling. Let $U_\phi(\cdot)$ and $U_\psi(\cdot)$ be single-qubit rotation encodings. The quantum q-deformed fusion is:

$$\text{Q-Fuse}_q = R_Y(\phi_g) \otimes R_Y(\psi_m) \cdot \exp\left(i\tilde{\theta} \cdot (Z \otimes Z - Z \otimes Z_{\text{swap}})\right), \tag{11}$$

where $Z \otimes Z$ is the standard Ising coupling, and $Z \otimes Z_{\text{swap}}$ denotes the swap-variant Ising term that breaks commutativity under qubit exchange.

Non-commutativity is verified via the standard SWAP-test:

$$F_{\text{SWAP}} = \left| \langle \tilde{\psi}_g, \psi_{\text{msg}} | \text{SWAP} | \psi_{\text{msg}}, \tilde{\psi}_g \rangle \right|^2 \neq 1 \quad (\tilde{\theta} \neq 0). \tag{12}$$

When $\tilde{\theta} \to 0$, the q-deformed phase vanishes, and Q-Fuse$_q$ degenerates to hybrid classical-quantum fusion.

Adaptive Update for $\tilde{\theta}$ via Parameter-Shift Rule In variational quantum circuits, gradients are computed via the two-point parameter-shift rule:

$$\frac{\partial \langle O \rangle}{\partial \tilde{\theta}} = \frac{1}{2} \left[\langle O \rangle|_{\tilde{\theta}+\frac{\pi}{2}} - \langle O \rangle|_{\tilde{\theta}-\frac{\pi}{2}} \right].$$

The twist angle update is only activated in the strong coherent regime:

$$\tilde{\theta}(t+1) = \tilde{\theta}(t) + \eta_\theta \cdot \nVdash_{\{\mathcal{C}^{q}_{\text{eff}}{}^{(g)} \geq \mathcal{C}^q_c\}} \cdot \left(\mathcal{C}^{q}_{\text{eff}}{}^{(g)} - \mathcal{C}^q_c \right) \cdot \frac{\partial \left(1 - F_{\text{SWAP}}\right)^2}{\partial \tilde{\theta}}. \tag{13}$$

The update magnitude is proportional to both the distance from the critical threshold and the quantum non-commutativity deficit $(1 - F_{\text{SWAP}})^2$.

3.8 Quantum Inter-Group Consistency Loss

Classical L2 distance is replaced by quantum state fidelity measured via SWAP-test:

$$\mathcal{L}^q_{\text{cons}} = -\sum_{(g,h)} W_{gh} \cdot F(\rho_g, \rho_h), \tag{14}$$

where $\rho_g = \text{Tr}_{\text{env}} \left(|\pi_g\rangle\langle\pi_g| \right)$ is the reduced density matrix of group g's policy, and $F(\cdot,\cdot)$ denotes quantum state fidelity.

3.9 Final Quantum Objective Function

The full loss function includes clipped quantum policy loss, fidelity-based consistency loss, quantum KL divergence, and von Neumann entropy regularization:

$$\begin{aligned} \mathcal{L}^{\text{Q-Topo-GRPO}} &= -\,\mathbb{E}\left[\min\left(r^q_t \hat{A}^g_{i,\text{quantum}}, \text{clip}(r^q_t, 1 \pm \epsilon) \hat{A}^g_{i,\text{quantum}} \right) \right] \\ &\quad + \lambda_{\text{cons}} \mathcal{L}^q_{\text{cons}} \\ &\quad + \beta D^q_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \\ &\quad - \gamma H_{\text{von}}(\rho_\theta), \end{aligned} \tag{15}$$

where $H_{\text{von}}(\rho) = -\text{Tr}(\rho \log \rho)$ is the von Neumann entropy, r^q_t is the quantum policy probability ratio, and D^q_{KL} denotes quantum relative entropy.

4 Quantum Training Algorithm

5 Theoretical Properties

- GRPO-aligned: no intra-group critics, advantage estimated purely from quantum reward statistics.
- Quantum speedup: quadratic speedup for neighborhood statistics via amplitude estimation.
- Entanglement-native: inter-group correlations modeled via Bell pairs without extra overhead.
- Topologically consistent: global quantum value guarantees cross-group coherence.
- NISQ-compatible: all modules adopt shallow variational quantum circuits.

Algorithm 1 Quantum Topology-GRPO Training

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Initialize high-level quantum policy $\pi_h^q$, group quantum policies $\{\pi_l^{g,q}\}$, quantum message encoders
Initialize global quantum value $V_{\text{global}}^q$ for topological calibration
Pre-distribute Bell pairs for all inter-group edges $E_{\text{inter}}$
Initialize $\mathcal{C}_c^q$ (fixed or learnable), initial $\tilde{\theta}^{(0)}$, learning rate $\eta_\theta$
while training not converged do
    Construct graph $G$ and split $E_{\text{intra}}, E_{\text{inter}}$
    Collect parallel quantum trajectories across all groups
    for each group $g$ do
        Measure $T_1^{(g)}, T_2^{(g)}$, compute Bell-pair fidelities $F_{gh}$
        Estimate quantum reward variance $\text{Var}_q(r_h)$ via amplitude estimation
        Calculate ${\mathcal{C}_{\text{eff}}^q}^{(g)}$ via Eq. (9)
        Perform Bell-pair quantum message passing to obtain $|\psi_{\text{msg}}\rangle_g$
        Compute enhanced state via Eq. (10)
        Compute quantum topological advantage via Eq. (8)
    end for
    Evaluate policy loss, fidelity loss, quantum KL divergence and von Neumann entropy
    Update quantum policy parameters via parameter-shift gradient descent
    if $\exists\, g$, ${\mathcal{C}_{\text{eff}}^q}^{(g)} \geq \mathcal{C}_c^q$ then
        Update $\tilde{\theta}$ via Eq. (13)
    end if
    Gradually decay $\lambda_{\text{cons}}$ for curriculum learning
end while
return $\pi_h^q, \{\pi_l^{g,q}\}$
```

6 Compatibility and Ablation Study

6.1 Backward Compatibility

When ${\mathcal{C}_{\text{eff}}^q}^{(g)} \to 0$ (decoherence-dominated regime) or $\tilde{\theta} \to 0$, Q-Fuse$_q$ degenerates to hybrid classical-quantum fusion, and the framework reduces to hybrid Topology-GRPO.

6.2 Ablation Design

1. Baseline: Original Quantum Topology-GRPO ($\mathcal{C}_{\text{eff}}^q$ and Q-Fuse$_q$ disabled).
2. $\mathcal{C}_{\text{eff}}^q$ only: Adaptive threshold enabled, always use hybrid fusion.
3. Full model: $\mathcal{C}_{\text{eff}}^q$ + Q-Fuse$_q$ + parameter-shift $\tilde{\theta}$ update.
4. Ablate non-commutativity: Fix $\tilde{\theta} = 0$.
5. Fixed $\mathcal{C}_c^q$ vs. learnable $\mathcal{C}_c^q$.

7 Implementation and Open Problems

Key implementation techniques for NISQ devices:

- Quantum topology-aware gradient clipping for noise robustness.
- Quantum running mean/std for reward normalization.
- Delayed entanglement synchronization for large-scale graphs.
- Curriculum learning for consistency regularization strength.

Open problems for future research:

1. Robustness under realistic NISQ device noise.

2. Optimization of inter-group entanglement topology.
3. Rigorous proof of quantum advantage over classical baselines.
4. Integration with quantum large language models (QLLMs).

8 Conclusion

Quantum Topology-GRPO upgrades the revised Topology-GRPO framework to the quantum domain while strictly preserving its core principles: no intra-group critics, reward-based advantage estimation, and global value for topological calibration only. The framework leverages quantum entanglement, amplitude estimation, and state fidelity to achieve efficient training and strong inter-group coordination, and is fully compatible with existing NISQ hardware.

The macro effective duality coupling $\mathcal{C}^{q}_{\text{eff}}$ and q-deformed Q-Fuse$_q$ are inspired by SED's duality philosophy but formulated as independent engineering modules. This work provides a theoretically grounded, hardware-friendly variational quantum hierarchical reinforcement learning framework for graph-structured decision tasks.